

Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons

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by

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Approval

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Declaration

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Soumen Kumar Mondal

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Abstract

This thesis investigates methodologies for identifying and utilizing language-specific neurons in multilingual large language models (LLMs) to improve performance on multilingual downstream tasks, particularly for low-resource languages. The research examines how language-specific information is represented in various LLM architectures, including Llama3.1, Bloomz, and Aya models. Techniques such as activation probability entropy and LoRA-based fine-tuning methods were applied to identify language-specific neurons using datasets like Wikipedia and XNLI. Intervention strategies, such as replacing neuron activations with language-specific statistical activations, were explored to determine their effect on task performance. Experiments involving neuron-specific fine-tuning and neuron freezing were also conducted to study the relationship between architectural modifications and language representation.

The study further analyzes perplexity distributions and neuron activation maps across different languages and models, providing insights into how multilingual datasets influence the internal mechanisms of LLMs. Experimental setups were developed to assess the effectiveness of fine-tuning and intervention methods for improving natural language inference (NLI) performance in low-resource languages without requiring direct training data. By addressing existing gaps in language-specific neuron identification, this work contributes to a better understanding of how LLMs can be tailored to achieve performance improvement across diverse languages. The results provide a deeper understanding of the interaction between LLM architecture and multilingual capabilities, particularly for underrepresented languages.

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Chapter 1

Introduction

1.1 Introduction

The emergence of large language models (LLMs) has transformed natural language processing (NLP), enabling remarkable advancements across a wide range of linguistic tasks. Despite their success, challenges persist in ensuring comparable performance for low-resource languages, which often lack sufficient representation in pretraining datasets. This thesis addresses these challenges by investigating the representation of language-specific information within LLMs and exploring methods to identify and utilize language-specific neurons. By focusing on multilingual tasks such as natural language inference (NLI), this research seeks to better understand how language-specific information is encoded and how it can be effectively utilized to improve performance, particularly for underrepresented languages.

The research builds on recent advancements in neuron analysis and fine-tuning techniques, using models such as Llama3.1, Bloomz, and Aya to explore the relationship between architectural choices, pretraining datasets, and language-specific representations. By employing methods such as activation probability entropy and LoRA-based fine-tuning, the study systematically identifies language-specific neurons and evaluates their role in downstream tasks. Through detailed experiments on layer-specific fine-tuning, freezing, and intervention strategies, the thesis aims to provide a comprehensive understanding of the mechanisms underlying multilingual capabilities in LLMs.

1.2 Motivation

In the era of multilingual LLMs, achieving consistent performance across diverse languages remains a significant challenge. High-resource languages benefit from extensive pretraining data and resources, leading to superior performance on downstream tasks, while low-resource

languages often lag behind. Addressing this imbalance is essential for ensuring that the advancements in NLP benefit speakers of all languages, regardless of the resources available for their linguistic representation.

Understanding the internal mechanisms of LLMs, specifically how they represent language-specific information, offers a way to addressing these challenges. By identifying language-specific neurons and their roles in encoding linguistic knowledge, we can develop targeted interventions to improve performance for low-resource languages. This research is driven by the need to close the performance gap, advance the understanding of multilingual model architectures, and contribute to the development of fair and inclusive NLP systems.

1.3 Problem Statement

While LLMs exhibit strong multilingual capabilities, their performance across languages is often inconsistent, with low-resource languages showing suboptimal results compared to high-resource counterparts. This disparity can be attributed to the lack of sufficient representation for low-resource languages during pretraining and the challenge of effectively isolating and utilizing language-specific information within these models. Existing methods for identifying language-specific neurons often rely on arbitrary thresholds or lack generalizability across different architectures and datasets. Moreover, the impact of architectural modifications, such as layer-specific fine-tuning or neuron freezing, on language-specific representation and downstream performance remains underexplored.

This research aims to address these limitations by developing systematic approaches to identify and analyze language-specific neurons in LLMs. By focusing on techniques such as neuron activation probability and LoRA-based fine-tuning, the study seeks to bridge the gap in understanding how language-specific information is encoded. The ultimate goal is to improve task performance for low-resource languages without requiring extensive annotated data, thereby contributing to the application of LLMs in multilingual settings.

Chapter 2

Literature Review

2.1 Language-Specific Neurons in Multilingual Models

A significant body of work has focused on understanding and isolating language-specific neurons within large language models (LLMs). The paper *Language-Specific Neurons in Multilingual Language Models* ([ArXiv:2404.02431](#)) investigates techniques for identifying neurons that are specific to certain languages and evaluates their impact on tasks such as machine translation and multilingual understanding. Building on this, our research extends the idea by evaluating alternative methods such as replacing median activations with layer-specific median or 75th percentile activations and exploring whether language-specific neuron sets vary across datasets, tasks, and model layers.

The reproduction of results from the *Language-Specific Regions* paper ([ArXiv:2402.16438](#)) further validated the connection between perplexity changes and language specificity in LLMs. The study provided insights into how replacing or interpolating neuron activations affects task performance, particularly for low-resource languages. These findings underline the importance of understanding neuron activity patterns for improving multilingual tasks. For instance, experiments demonstrated that full replacement of language-specific neurons may degrade performance, while linearly interpolating activation values can balance task performance and neuron specificity.

2.2 Mechanistic Interpretability and Multilinguality

The broader context of neuron analysis in multilingual LLMs intersects with mechanistic interpretability, as explored in works such as the *SoLU Activation Function* ([SoLU paper](#)) and *Tracing the Roots of Facts in Multilingual Models* ([ArXiv:2403.05189](#)). The former highlights methods to reduce polysemanticity in neurons, which could aid in isolating neurons responsible for specific linguistic features. Meanwhile, the latter paper delves into probing techniques for

factual association in multilingual models, suggesting that neuron activity patterns can vary significantly depending on the dataset and task.

This thesis builds on these foundational ideas by experimenting with probing methods such as L1 and L2 norms and designing novel interventions in neuron activations. The literature also emphasizes how architectural changes, such as adding adapter layers or modifying attention heads, can enhance model performance on multilingual tasks. The exploration of neuron activity patterns for cross-lingual transfer, fact probing, and multilingual perplexity maps adds depth to the understanding of language representation in LLMs.

2.3 Applications to Low-Resource Languages

Addressing performance disparities for low-resource languages is a recurring theme across the literature. Papers such as *Locating and Editing Factual Associations in GPT* and *Discovering Salient Neurons in Deep NLP Models* highlight methodologies for identifying and modifying neurons to improve downstream task performance. This aligns with our research objectives, particularly in the context of improving natural language inference (NLI) and translation tasks for low-resource languages. Techniques such as activation replacement, layer freezing, and fine-tuning specific neurons have shown promise in achieving consistent task performance across diverse linguistic settings.

Chapter 3

Methodology

3.1 Definition of Neuron

In the context of multi layer perceptrons (MLPs) in large language models (LLMs), a *neuron* refers to a unit of computation within the MLP module. Specifically, consider the i^{th} layer of a Transformer architecture, where the input is a hidden state vector $\mathbf{H}^{i-1} \in \mathbb{R}^{T \times d}$, derived from the previous layer.

Hidden State Representation: The output of the multi-head self-attention (MHA) mechanism serves as input to the MLP and is given as:

$$\hat{\mathbf{H}}^i = \text{Attn}(\mathbf{H}^{i-1}\mathbf{W}_q^i, \mathbf{H}^{i-1}\mathbf{W}_k^i, \mathbf{H}^{i-1}\mathbf{W}_v^i) \cdot \mathbf{W}_o^i, \quad (3.1)$$

where $\mathbf{W}_q^i, \mathbf{W}_k^i, \mathbf{W}_v^i, \mathbf{W}_o^i \in \mathbb{R}^{d \times d}$ are the trainable parameters of the self-attention module, and $\text{Attn}(\cdot)$ denotes the scaled dot-product attention operation.

MLP Transformation: The MLP module takes $\hat{\mathbf{H}}^i$ as input and transforms it as follows:

$$\mathbf{h}^i = \text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \cdot \mathbf{W}_2^i, \quad (3.2)$$

where $\mathbf{W}_1^i \in \mathbb{R}^{d \times 4d}$ and $\mathbf{W}_2^i \in \mathbb{R}^{4d \times d}$ are learnable projection matrices, and $\text{act_fn}(\cdot)$ represents a non-linear activation function such as GELU or SiLU.

Neuron: Within the MLP, the j^{th} neuron of the i^{th} layer corresponds to the j^{th} row of the intermediate activation matrix:

$$\mathbf{h}_j^i = \text{act_fn}((\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j), \quad (3.3)$$

where $(\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j$ represents the pre-activation value of the j^{th} neuron.

Generalized Neuron Formulation: Recent architectures such as LLaMA utilize Gated Linear Units (GLU) to enhance MLP performance, modifying the MLP computation as:

$$\mathbf{h}^i = [\text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (3.4)$$

where $\mathbf{W}_3^i \in \mathbb{R}^{d \times 4d}$ is an additional projection matrix, and $\otimes$ denotes element-wise multiplication. This formulation introduces gating to the MLP, allowing for more expressive transformations. Figure 3.1 shows the architecture of the MLP module.

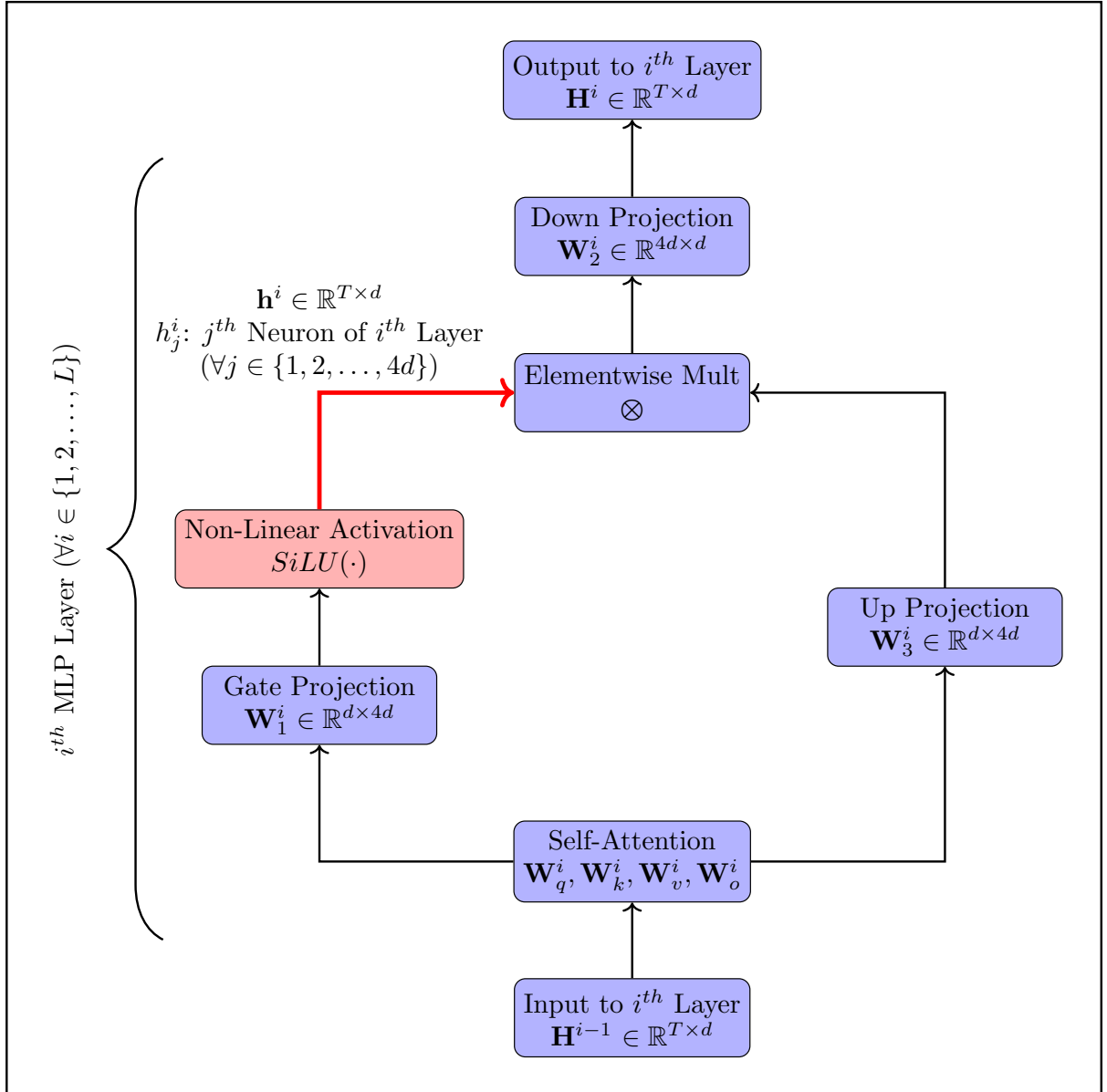


FIGURE 3.1: Architecture of MLP module in LLaMA model

3.2 Dataset for Neuron Identification

For the purpose of identifying language-specific neurons, we utilize a subset of the Wikipedia dataset spanning 15 languages:

$$\{\text{en, fr, es, vi, id, ja, zh, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

Each language’s dataset is preprocessed and passed through pretrained large language models (LLMs). The activations are then collected from the MLP modules of the models.

Dataset Description: The Wikipedia dataset for each language contains a sequence of tokenized text data. Each tokenized sequence has a maximum length of $T_{\max} = 512$. For every sequence, we compute the activations from the MLP module, followed by averaging the activations at each neuron position across all sequences. This provides a language-specific profile for every neuron (i, j) in the model.

Activation Collection: Given a pretrained LLM, let the MLP activation for the i^{th} layer and the j^{th} neuron in a specific language be represented by $h_{i,j}^t \in \mathbb{R}$, where t indexes the token positions in a sequence of length T . The average activation of a neuron across all token positions for a given sequence is computed as:

$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.5)$$

where $h_{i,j}^t$ is the neuron activation corresponding to the t^{th} token.

Language-Specific Activations: For each language l in the dataset, we compute the average activation of each neuron (i, j) over all sequences S_l in the dataset:

$$h_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3.6)$$

where $\bar{h}_{i,j}^l$ represents the average activation of the j^{th} neuron in the i^{th} layer for language l , and $|S_l|$ denotes the number of sequences for language l .

3.3 Statistics of Neuron Activation

The statistics of neuron activation are precomputed to analyze language-specific patterns in the MLP layers of the pretrained large language models (LLMs). These statistics include mean, standard deviation, and quantiles (e.g., 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles) of the neuron activations for each language in the dataset. The computations are performed across all layers and neurons within the MLP modules.

Activation Mean and Standard Deviation: The mean activation of a neuron (i, j) in the t^{th} token position is computed as:

$$\mu_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.7)$$

where $h_{i,j}^t$ is the activation of the (i, j) neuron for the t^{th} token, and T is the total number of tokens in a sequence. Similarly, the standard deviation of the activations is calculated as:

$$\sigma_{i,j} = \sqrt{\frac{1}{T} \sum_{t=1}^T (h_{i,j}^t - \mu_{i,j})^2}. \quad (3.8)$$

Quantiles of Activation: Quantile-based statistics provide insights into the distribution of activations for each neuron. The q -quantile activation value, denoted as P_q , is defined as:

$$P_q(h_{i,j}) = \inf \{x \in \mathbb{R} : F_{h_{i,j}}(x) \geq q\}, \quad (3.9)$$

where $F_{h_{i,j}}(x)$ is the cumulative distribution function (CDF) of the activations for neuron (i, j) . Commonly computed quantiles include:

$$P_{50}, P_{75}, P_{90}, P_{95}, P_{25}, P_{10}, P_5.$$

Language-Specific Aggregation: For a given language l and all sequences in the dataset, the mean and other statistics are aggregated across all token positions and batches. Let S_l denote the set of sequences for language l , and B represent the batch size. The aggregated mean activation for neuron (i, j) is computed as:

$$\bar{\mu}_{i,j}^l = \frac{1}{|S_l|} \sum_{b=1}^B \sum_{s \in S_l} \mu_{i,j}^{(s,b)}, \quad (3.10)$$

where $\mu_{i,j}^{(s,b)}$ is the mean activation of neuron (i, j) for the s^{th} sequence in the b^{th} batch.

3.4 Methods for Neuron Identification

3.4.1 LAPE

Language Activation Probability Entropy (LAPE) is a metric designed to identify neurons that exhibit language-specific behavior across multiple languages. This metric quantifies the entropy of normalized activation probabilities for each neuron across the given set of languages.

Step 1: Compute Activation Probabilities. For a pretrained large language model with L layers and $4d$ neurons per layer, the activation of a neuron (i, j) in response to a sequence $X^l \in \mathbb{R}^{N \times T}$ (where N is the number of sequences, and T is the number of tokens per sequence)

for a language l is denoted as $h_{i,j}^l$. The unnormalized activation probability for neuron (i, j) is given by:

$$p_{i,j}^l = \mathbb{E} [\mathbb{I} (h_{i,j}^l > 0)] , \quad (3.11)$$

where $\mathbb{I}$ is the indicator function that returns 1 if the activation $h_{i,j}^l$ is greater than zero, and 0 otherwise. The expectation is computed over all tokens in all sequences for language l .

Step 2: Normalize Activation Probabilities. The normalized activation probability is computed across k languages to ensure a valid probability distribution:

$$P_{i,j}^l = \frac{p_{i,j}^l}{\sum_{l=1}^k p_{i,j}^l}, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.12)$$

Step 3: Compute Language Activation Probability Entropy. The entropy of the normalized activation probabilities for each neuron (i, j) is computed as:

$$\text{LAPE}(i, j) = - \sum_{l=1}^k P_{i,j}^l \log P_{i,j}^l, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.13)$$

This measures the uncertainty in the activation distribution of a neuron across the k languages. Figure 3.2 shows the procedure of computing LAPE for a neuron.

Step 4: Apply Threshold and Select Neurons. To identify neurons with significant language-specific behavior, a threshold is defined based on the 95th percentile of the unnormalized activation probabilities:

$$\text{Threshold} = \text{Quantile}_{0.95} (p_{i,j}^l \mid \forall i, j, l). \quad (3.14)$$

A neuron (i, j) is considered for language-specific analysis only if:

$$\exists l \in \{1, \dots, k\} \text{ such that } p_{i,j}^l > \text{Threshold}. \quad (3.15)$$

Step 5: Select Top Neurons Based on LAPE. The top m neurons with the lowest LAPE values are selected for language-specific analysis, where m is defined as:

$$m = \frac{0.01 \times 4d \times L}{100}. \quad (3.16)$$

The top m neurons are identified by sorting the LAPE values and selecting the indices of the m smallest values:

$$\text{Top}_m(\text{Lang_Set}) = \text{argsort_ascending}_m \{ \text{LAPE}(i, j) \mid \forall i, j, \exists l \text{ s.t. } p_{i,j}^l > \text{Threshold} \}. \quad (3.17)$$

This ensures that only the most language-specific neurons are retained for further interpretation.

Step 6: Identify Language-Specific Neuron Sets. For each neuron (i, j) in the selected set, the language-specific set is defined as:

$$\text{Lang_Spec_Set}(i, j) = \{l \mid p_{i,j}^l > \text{Threshold}, (i, j) \in \text{Lang_Set}\}. \quad (3.18)$$

A neuron is considered specific to a subset of languages if:

$$\text{Lang_Spec_Set}(i, j) \subseteq \{1, 2, \dots, k\}. \quad (3.19)$$

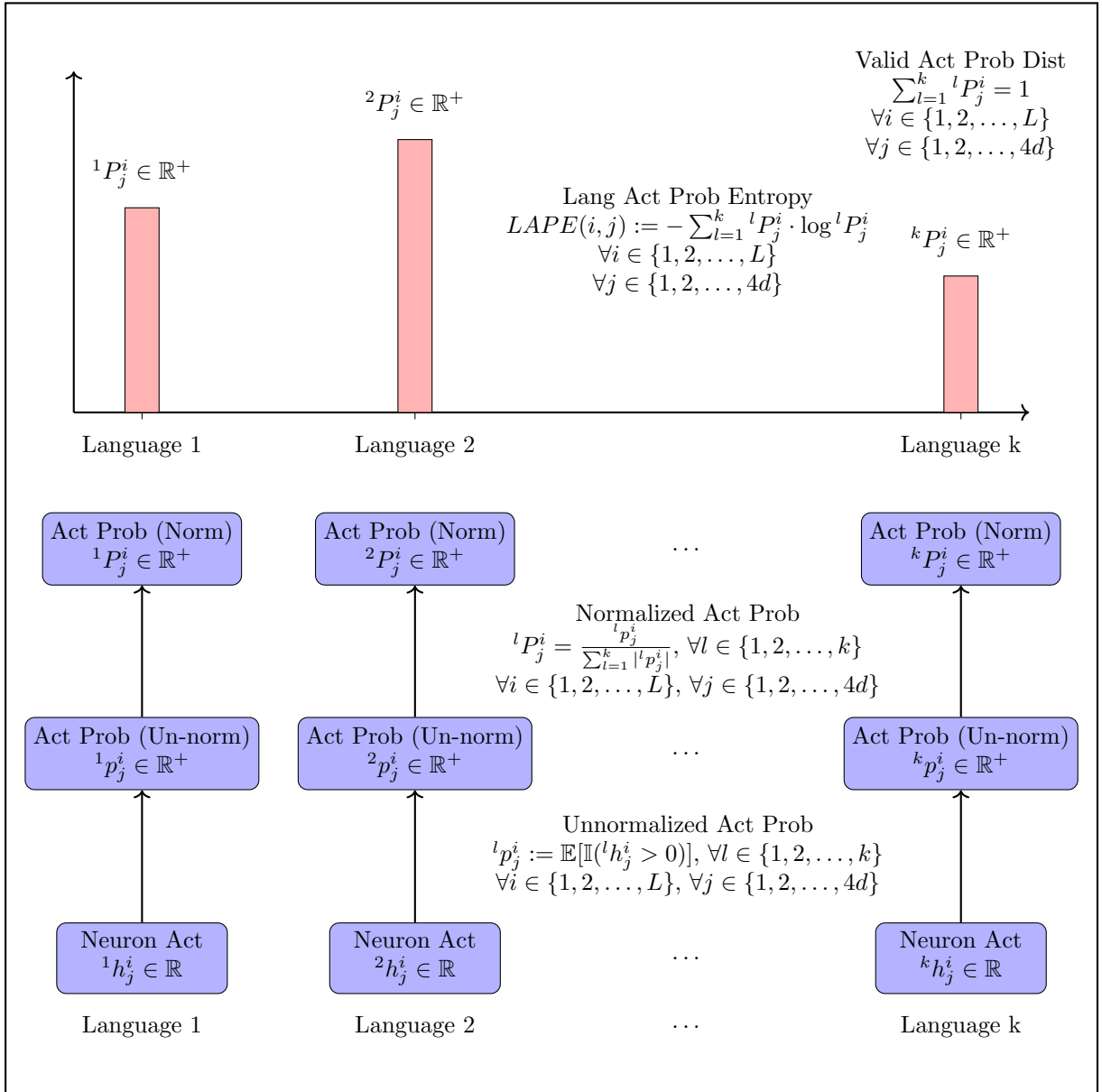


FIGURE 3.2: Procedure of computing LAPE for a neuron.

3.4.2 Activation Based Methods

Absolute Activation: The absolute activation metric measures the mean absolute activation of each neuron (i, j) in the i^{th} layer across all tokens in a sequence for a language l . This is computed as:

$$r_{i,j}^l = \mathbb{E} [|h_{i,j}^l|], \quad (3.20)$$

where $h_{i,j}^l$ is the activation of neuron (i, j) for language l , and the expectation is taken over all tokens and sequences. This metric identifies neurons with consistently high absolute activations, which are indicative of their overall importance in representing language features.

Activation Probability Zero: The activation probability zero metric measures the fraction of times a neuron (i, j) has a positive activation for a language l . It is defined as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.21)$$

where $\mathbb{I}$ is the indicator function that returns 1 if $h_{i,j}^l > 0$, and 0 otherwise. This metric captures the frequency of a neuron's activation, providing insights into its contribution to language-specific representations.

Activation Probability Mean: The activation probability mean measures the probability of a neuron's activation exceeding the mean activation value for its layer, $\mu_{i,j}^l$, for a language l . It is computed as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > \mu_{i,j}^l)], \quad (3.22)$$

where $\mu_{i,j}^l$ represents the average activation of neuron (i, j) in the i^{th} layer for language l . This metric identifies neurons with activations that are consistently above the average, highlighting neurons that are highly responsive to language-specific features.

Activation Probability 90p: The activation probability 90p measures the probability of a neuron's activation exceeding the 90th percentile activation value for its layer, $\text{P90}_{i,j}^l$, for a language l . This is given by:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > \text{P90}_{i,j}^l)], \quad (3.23)$$

where $\text{P90}_{i,j}^l$ represents the 90th percentile of the activations of neuron (i, j) in the i^{th} layer for language l . This metric identifies neurons that exhibit rare but significant activations, which may correspond to specialized processing for certain linguistic features.

Language-specific neurons are identified based on activation values computed across different languages. For each language, activations are collected, and neurons that exhibit the highest relevance are selected using the following process.

Step 1: Compute Activation Relevance. For each language $l \in \{1, 2, \dots, k\}$, compute the activation relevance values for all neurons (i, j) in the pretrained model. Let the relevance for neuron (i, j) in the i^{th} layer for language l be denoted as $r_{i,j}^l$. This relevance is calculated using one of the activation probability methods, such as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.24)$$

or other methods such as absolute activation or percentile-based thresholds, depending on the desired analysis.

Step 2: Aggregate Relevance Across Layers and Neurons. For each language l , the computed relevance values $r_{i,j}^l$ are aggregated across all layers and neurons. The total number of layers is denoted by L , and the dimensionality of the neurons in each layer is $4d$. The relevance tensor for language l is represented as:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\}, j \in \{1, \dots, 4d\}\}. \quad (3.25)$$

Step 3: Select Top Neurons. For each language l , the top m neurons with the highest relevance values are selected. The value of m is determined as a small fraction of the total number of neurons in the model:

$$m = \frac{0.125 \times L \times 4d}{100}. \quad (3.26)$$

The top m neurons are identified by sorting the relevance values in R^l and selecting the indices of the m largest values:

$$\text{Top}_m(R^l) = \text{argsort_descending}_m \{r_{i,j}^l \mid \forall i, j\}. \quad (3.27)$$

Step 4: Map Neurons to Layers and Positions. The indices of the top m neurons are mapped back to their corresponding layers and positions. For each language l , the selected neurons are represented as:

$$N^l = \{(i, j) \mid (i, j) \in \text{Top}_m(R^l)\}, \quad (3.28)$$

where N^l is the set of top m neurons for language l .

This process ensures that neurons with the highest language-specific relevance are identified for each language. These neurons can then be analyzed further to understand their behavior and contribution to language modeling tasks.

3.4.3 Task Neurons for a Language

In this method, we introduce an alternative method based on task-specific neurons computed from the task specific dataset (for example XNLI dataset). Unlike language-specific neurons, which are computed using activation patterns derived from Wikipedia sentences, the task neurons are computed by processing the training sentences from the task dataset for each respective language. This modification allows the identification of neurons that are not only language-specific but also task-specific, capturing finer granularity of representations relevant to the classification task.

Activation of Task-Specific Neurons: The activation pattern for task-specific neurons is derived as:

$$h_{ij}^{\text{task}} = \text{stat}_{ij}^{\text{task}}, \quad \forall \text{stat} \in \{\mu, p_{90}, p_{75}, p_{50}, p_{25}, p_{10}, p_5\}, \quad (3.29)$$

where:

- h_{ij}^{task} represents the activation for the i -th layer and j -th neuron under task-specific conditions.
- $\text{stat}_{ij}^{\text{task}}$ denotes the statistical activation computed over all task specific training sentences in a given language.
- The set $\{\mu, p_{90}, p_{75}, p_{50}, p_{25}, p_{10}, p_5\}$ represents the mean, 90th percentile, 75th percentile, and other quantile thresholds used to analyze neuron activation levels.

Once the activation is computed for a task, then we can apply the LAPE or any activation based methods to compute the task specific neurons.

3.5 Neuron Fine-Tuning using LoRA for a Task

LoRA (Low-Rank Adaptation): LoRA introduces a mechanism to fine-tune a subset of parameters in neural networks, focusing on reducing computational overhead. For our application, we utilize LoRA to specifically fine-tune targeted neurons in the MLP layer while freezing others. This method ensures that the pretrained parameters W_{pt} remain untouched, and ΔW represents the modifications applied to the network during fine-tuning.

Fine-Tuning with LoRA AB Decomposition. The matrix ΔW is expressed as:

$$\Delta W = A \cdot B, \quad (3.30)$$

where $A \in \mathbb{R}^{d_{in} \times r}$ and $B \in \mathbb{R}^{r \times d_{out}}$ are low-rank matrices. The rank r is typically much smaller than d_{in} or d_{out} , reducing the number of trainable parameters. Masks are applied to A and B to control the fine-tuning of specific neurons:

$$A = \text{masked_A} \cdot A_t, \quad B = \text{masked_B} \cdot B_t, \quad (3.31)$$

where masked_A and masked_B are binary masks, and A_t and B_t are trainable parameters.

Description of Masked Matrices:

The binary masks masked_A and masked_B are defined to control the fine-tuning process by selectively freezing specific neurons.

$$\text{masked_A} \in \mathbb{R}^{d_{in} \times r}, \quad \text{masked_B} \in \mathbb{R}^{r \times d_{out}},$$

where d_{in} and d_{out} are the input and output dimensions of the layer, and r is the rank of the LoRA decomposition.

Masked Matrix masked_A : masked_A is defined as:

$$[\text{masked_A}]_{ij} = 1, \quad \forall i \in \{1, \dots, d_{in}\}, \forall j \in \{1, \dots, r\}. \quad (3.32)$$

This indicates that no masking is applied to the rows of A , and all elements of masked_A are set to 1.

Masked Matrix masked_B : For masked_B , specific columns corresponding to frozen neurons are set to 0. For a neuron index $j_f \in \text{Frozen Neurons}$, the mask is defined as:

$$[\text{masked_B}]_{ij} = \begin{cases} 0, & \text{if } j = j_f, j_f \in \text{Frozen Neurons}, \\ 1, & \text{otherwise.} \end{cases} \quad (3.33)$$

Final Masking Condition: The effective masked matrices for A and B are:

$$A = \text{masked_A} \cdot A_t, \quad B = \text{masked_B} \cdot B_t, \quad (3.34)$$

where $A_t \in \mathbb{R}^{d_{in} \times r}$ and $B_t \in \mathbb{R}^{r \times d_{out}}$ are the trainable LoRA matrices.

This masking ensures that updates to the frozen neurons are completely suppressed, while other neurons remain trainable during fine-tuning.

Frozen Neurons. To freeze specific neurons, columns of B corresponding to the frozen neurons are masked with zeros. For neuron indices j in the set of frozen neurons, we ensure:

$$\Delta W_{i,j} = 0, \quad \forall i. \quad (3.35)$$

This prevents any updates to these neurons during fine-tuning.

Final Output. The fine-tuned output of the MLP layer is given by:

$$y = x \cdot W + x \cdot \Delta W, \quad (3.36)$$

where W is the original weight matrix, and ΔW introduces targeted updates through LoRA.

Targeted Language Neurons. Language-specific neurons are identified and fine-tuned using LoRA. This targeted approach ensures that only the relevant neurons for a particular language are adapted while the others remain unchanged. For the frozen neurons, ΔW remains zero:

$$\Delta W_{i,j} = 0, \quad \text{if } j \in \text{Frozen Neurons}. \quad (3.37)$$

Additional Fine-Tuning. Besides fine-tuning the MLP layer, the attention mechanisms (q, k, v, o projections) and the task-specific classifier head are always fine-tuned using LoRA. This ensures the overall task-specific performance is preserved while leveraging the language neurons for targeted adaptation. Figure 3.3 shows the procedure of finetuning target neurons using LoRA.

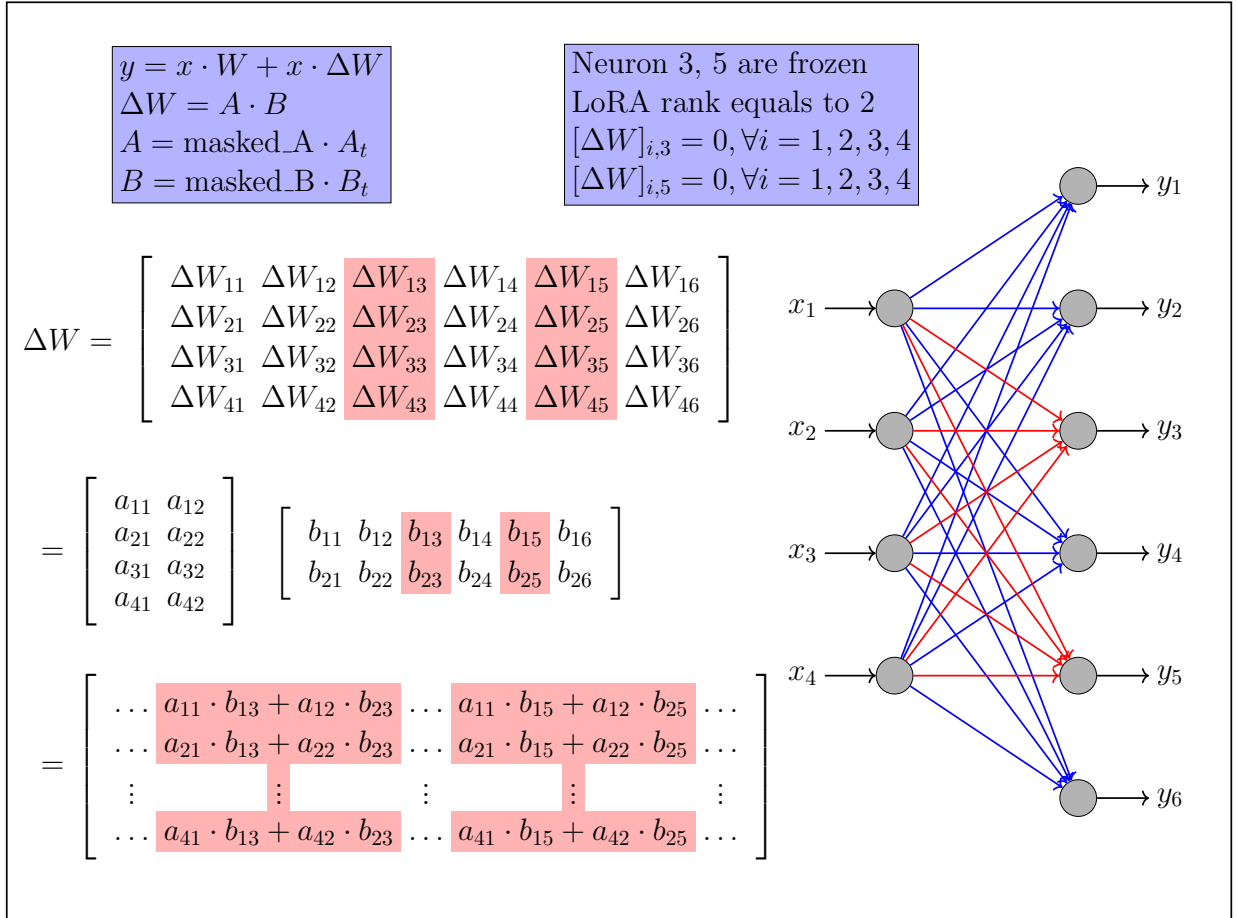


FIGURE 3.3: LoRA fine tuning method for target neurons.

Chapter 4

Experiments

4.1 Task and Dataset: XNLI

The Cross-lingual Natural Language Inference (XNLI) task is a benchmark designed to evaluate the capability of large language models (LLMs) to perform reasoning across multiple languages. The goal is to classify the relationship between a premise and a hypothesis into one of three categories: *entailment*, *neutral*, or *contradiction*. This task tests the ability of the model to generalize semantic relationships beyond the English language, making it a challenging benchmark for multilingual models.

Dataset Overview: The XNLI dataset contains annotated sentence pairs across 15 languages. In our experiments, we focus on English (*en*) as the source language and Vietnamese (*vi*), Hindi (*hi*) and Urdu (*ur*) as target languages. The dataset for each language can be represented as:

$$\mathcal{D}_l = \{(x_p^i, x_h^i, y^i) \mid x_p^i, x_h^i \in \mathcal{X}_l, y^i \in \{0, 1, 2\}\}, \quad (4.1)$$

where x_p^i and x_h^i denote the premise and hypothesis sentences in language l , and y^i is the corresponding label indicating the relationship.

Tokenization and Preprocessing: Each sentence pair (x_p^i, x_h^i) is concatenated with a special token to form the input:

$$x^i = x_p^i + [\text{EOS}] + x_h^i, \quad (4.2)$$

where $[\text{EOS}]$ represents the end-of-sequence token. The concatenated input is tokenized into a sequence of tokens $\{t_1, t_2, \dots, t_T\}$ with a maximum sequence length $T = 512$.

Train and Test Splits: The dataset is divided into training and testing subsets for each language:

$$\mathcal{D}_l = \mathcal{D}_l^{\text{train}} \cup \mathcal{D}_l^{\text{test}}, \quad \mathcal{D}_l^{\text{train}} \cap \mathcal{D}_l^{\text{test}} = \emptyset. \quad (4.3)$$

For the experiments, a fraction $\mathcal{F} \subset \mathcal{D}_l^{\text{train}}$ is used to train the model. The size of $\mathcal{F}$ is determined as:

$$|\mathcal{F}| = \text{frac} \times |\mathcal{D}_l^{\text{train}}|, \quad (4.4)$$

where frac is the fraction of the training dataset used.

Multilingual Challenges: In the XNLI task, transferring knowledge from English to other languages like Vietnamese and Hindi requires the model to overcome both semantic and structural differences in language. This transfer is tested by evaluating the zero-shot performance on the target languages without directly training on their data. The objective of this task is to fine-tune LLaMA3, a large language model, on the XNLI dataset to improve cross-lingual inference capabilities. The experiments are designed to test the effectiveness of various fine-tuning setups and analyze the role of language-specific neurons in improving performance across multiple languages.

4.2 Baseline Experiments

The baseline experiments establish a foundation for evaluating the performance of different fine-tuning setups on the XNLI task. We include two baselines: zero-shot evaluation of a model fine-tuned on English (*en*) XNLI data and direct fine-tuning on the target language (*vi*) XNLI data. These baselines provide a comparative framework to assess the impact of fine-tuning specific language neurons using LoRA.

Zero-Shot Baseline: The first baseline involves fine-tuning the pretrained model on the English XNLI dataset. Let $\mathcal{M}_{\text{en}}$ denote the model fine-tuned on English data. The zero-shot evaluation involves testing $\mathcal{M}_{\text{en}}$ on Vietnamese XNLI datasets without further training. The evaluation metric, accuracy, is defined as:

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(\hat{y}^i = y^i)}{N}, \quad (4.5)$$

where N is the total number of test examples, y^i is the ground truth label, $\hat{y}^i$ is the predicted label, and $\mathbb{I}(\cdot)$ is the indicator function.

Direct Fine-Tuning Baseline: The second baseline involves directly fine-tuning the pretrained model on the Vietnamese XNLI dataset. Let $\mathcal{M}_{\text{vi}}$ denote the model fine-tuned on Vietnamese data. This baseline establishes an upper bound for performance on the Vietnamese XNLI task, as the model is explicitly trained on the target language data. We again use the accuracy as evaluation metric.

4.3 Experiment Goals

The primary objective of this work is to improve the zero-shot performance of the model on the target language, such as Vietnamese (*vi*), without utilizing any target language training data. This constraint is critical to ensure that the model remains agnostic to the specific characteristics of the target language during training, thereby simulating a real-world cross-lingual transfer learning scenario.

Baseline Comparison: In this setup, we consider two baselines:

- **Baseline 1: Zero-Shot Performance on *vi* using English Training Data.** This baseline involves fine-tuning the pretrained model on the English (*en*) XNLI dataset ($\mathcal{M}_{\text{en}}$) and evaluating its zero-shot performance on the Vietnamese (*vi*) test set. This represents the starting point of cross-lingual transfer without any modifications to the model architecture or targeted interventions.
- **Baseline 2: Performance on *vi* using Direct Fine-Tuning on *vi* Data.** This baseline involves directly fine-tuning the pretrained model on the Vietnamese (*vi*) XNLI dataset ($\mathcal{M}_{\text{vi}}$). It provides an upper bound on performance since it leverages target language training data.

Objective Constraints: While the ultimate goal is to improve the zero-shot performance on the *vi* test set, it is imperative to operate under the following constraints:

1. No target language (*vi*) training data can be used during the fine-tuning process.
2. Only the English (*en*) XNLI training data is available for model training and intervention.
3. The interventions must be limited to model surgery techniques, such as fine-tuning specific neurons or modifying activation patterns.

Target Metric: The improvement in zero-shot performance is quantified as:

$$\Delta_{\text{ZS}} = \text{Accuracy}(\mathcal{M}_{\text{experiment}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}), \quad (4.6)$$

where $\mathcal{M}_{\text{experiment}}$ refers to the model modified through various intervention strategies, and $\mathcal{M}_{\text{en}}$ refers to the baseline model fine-tuned on *en*.

Goal: The overarching goal is to maximize Δ_{ZS} , thereby surpassing the performance of Baseline 1 ($\mathcal{M}_{\text{en}}$) through innovative techniques such as:

- Fine-tuning language-specific neurons identified for the target language (*vi*) using only the *en* training data.
- Utilizing statistical activation patterns (*e.g.*, *mean*, *quantile-based thresholds*) to intervene in neuron activations during inference.
- Applying LoRA (Low-Rank Adaptation) to selectively fine-tune specific components such as *qkv* projections and classification heads while freezing other parts of the model.

Limitations: By design, the performance of any intervention cannot exceed Baseline 2 ($\mathcal{M}_{\text{vi}}$), as that baseline uses direct training data from the target language. However, our approach aims to close the gap between Baseline 1 ($\mathcal{M}_{\text{en}}$) and Baseline 2 ($\mathcal{M}_{\text{vi}}$) as much as possible through zero-shot strategies.

4.4 Targeted Neuron Fine-tuning Experiment

This section outlines the experimental setups designed to improve the zero-shot performance of the model on the Vietnamese (*vi*) XNLI dataset. Four configurations are explored, each using different strategies to fine-tune the model, while adhering to the constraint of using only English (*en*) training data during fine-tuning. The experiments aim to utilize targeted interventions to improve performance by selectively tuning neurons and applying LoRA-based adaptations.

Setup A: Standard Fine-Tuning with LoRA. In this setup, the pretrained model is fine-tuned on the English (*en*) XNLI dataset using LoRA. The fine-tuning is applied to the query (q), key (k), value (v), and output (o) projection matrices ($W_q^i, W_k^i, W_v^i, W_o^i$) and the classification head. The resulting model is evaluated on the Vietnamese (*vi*) dataset in a zero-shot setting. Additionally, the *vi*-specific activations are substituted by statistical activation patterns such as mean (μ), 90th percentile, and others, defined as:

$$\hat{h}_j^{i,vi} = \text{stat}_j^{i,vi}, \quad \forall \text{stat} \in \{\text{mean}, 90p, 95p, 75p, 5p, 10p, 25p, 0\}, \quad (4.7)$$

where $\hat{h}_j^{i,vi}$ represents the substituted activation for *vi* language for neuron (i, j) in the MLP block.

Setup B: Fine-Tuning with LoRA and English Neurons. This setup builds on Setup A by incorporating fine-tuning of English-specific neurons identified in the MLP block. Let $\mathcal{N}_{\text{en}}$ denote the set of English-specific neurons. During fine-tuning, the LoRA modifications (ΔW) are applied to both the query-key-value-output projections and the English neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{en}}. \quad (4.8)$$

This targeted approach allows the model to specialize in English while preserving the generalization capacity of other neurons.

Setup C: Fine-Tuning with LoRA and Target Language Neurons. In this configuration, the focus shifts to fine-tuning target language such as Vietnamese-specific neurons ($\mathcal{N}_{\text{vi}}$). During fine-tuning, the LoRA modifications (ΔW) are applied to both the query-key-value-output projections and the Vietnamese neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{vi}}. \quad (4.9)$$

This ensures that only the Vietnamese-specific neurons are fine-tuned while others remain untouched.

Setup D: Fine-Tuning with LoRA and Both English & Target Language Neurons. This setup combines the interventions from Setup B and Setup C, targeting both English and Vietnamese neurons. Let $\mathcal{N}_{\text{en,vi}} = \mathcal{N}_{\text{en}} \cup \mathcal{N}_{\text{vi}}$ represent the union of English and Vietnamese neurons. During fine-tuning, the LoRA modifications (ΔW) are applied to both the query-key-value-output projections and both the English & Vietnamese neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{en,vi}}. \quad (4.10)$$

This strategy allows the model to simultaneously adapt to English and Vietnamese language-specific representations.

Comparative Analysis: After fine-tuning, the zero-shot performance on Vietnamese (*vi*) XNLI data is measured using the accuracy metric. The performance improvements achieved through the setups are quantified by:

$$\Delta_{\text{ZS}} = \text{Accuracy}(\mathcal{M}_{\text{setup}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}), \quad (4.11)$$

where $\mathcal{M}_{\text{setup}}$ refers to the model fine-tuned using one of the experimental setups. These improvements are compared against the baselines to evaluate the efficacy of the targeted neuron fine-tuning strategies.

4.5 Targeted Neuron Fine-Tuning Configurations

Training Overview: The training of the XNLI task was conducted using the LoRA-based fine-tuning strategy, leveraging the Llama3.1 model. The experimental setup includes both English and multilingual datasets, with targeted interventions in language-specific neurons. The primary objective was to fine-tune the model while freezing specific subsets of neurons based on language-specific activations.

Model and Quantization: The Llama3 model was employed with 4-bit quantization, utilizing the nf4 quantization type for memory efficiency. The compute precision for quantization was set to bfloat16 with double quantization enabled. This configuration ensures minimal computational overhead without compromising the model's performance.

Dataset: The dataset used for training and evaluation was the **XNLI Dataset**. The dataset was tokenized using the pre-trained tokenizer corresponding to the Llama3 model, with a maximum context length of 512 tokens. The training dataset utilized 25% of the available data, while 10% was reserved for evaluation.

Training Hyperparameters:

- **Epochs:** Training was conducted for a single epoch, divided into steps calculated based on the batch size and dataset size.
- **Batch Size:** A batch size of 8 was used for both training and evaluation, with gradient accumulation steps set to 1.
- **Learning Rate:** The initial learning rate was set to 1×10^{-5} , with a linear warm-up scheduler for the first 1% of steps.
- **Optimization:** The optimizer used was AdamW with $\beta_1 = 0.95$, $\beta_2 = 0.999$, weight decay of 0.1, and a gradient clipping norm of 10.0.
- **LoRA Configuration:** The LoRA rank was set to 8, and the scaling factor (α) was set to 16.

Chapter 5

Analysis and Results

5.1 LAPE Neuron Analysis

5.1.1 Language Sets

In this section, we define the different language sets used for evaluating the Language Activation Probability Entropy (LAPE) in the context of our experiments. The language sets are constructed to capture varying linguistic and typological characteristics across a diverse group of languages. The experimental results on these sets aim to analyze the impact of multilinguality on language-specific neuron activations.

Set 1: Core Languages. This set includes the core languages commonly supported by pretrained language models:

$$\text{Set 1} = \{\text{en, fr, es, vi, id, ja, zh}\}.$$

These languages span across major linguistic families and represent both high-resource and typologically distinct languages.

Set 2: Extended Core with Indian Languages. To incorporate a mix of Western and Indian languages, this set is defined as:

$$\text{Set 2} = \{\text{en, fr, vi, zh, bn, hi, te, mr}\}.$$

Here, bn (Bengali), hi (Hindi), te (Telugu), and mr (Marathi) represent Indian languages, providing an interesting contrast to the Western languages in the set.

Set 3: Indian Subset. This set focuses exclusively on Indian languages:

$$\text{Set 3} = \{\text{bn, hi, ta, te, mr, ur, kn, ml, pa}\},$$

where ta (Tamil), ur (Urdu), kn (Kannada), ml (Malayalam), and pa (Punjabi) are included to cover a broader typological spectrum of Indian languages.

Set 4: Core and Extended Languages. Combining the core and extended sets, this set is defined as:

$$\text{Set 4} = \{\text{en, fr, es, vi, id, ja, zh, bn, hi, te}\}.$$

This allows for the comparison of results across a comprehensive multilingual setup.

Set 5: All Languages. This set includes all the languages in the experimental framework:

$$\text{Set 5} = \{\text{en, fr, es, vi, id, ja, zh, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

Set 6: Indian Language Dominant. This set focuses on Indian languages while retaining a few globally dominant ones:

$$\text{Set 6} = \{\text{en, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

The diversity of the language sets ensures a thorough evaluation of language-specific neuron behavior. By analyzing LAPE across these sets, we aim to uncover insights into the multilingual capabilities of the pretrained model and the role of specific languages in driving neuron activations.

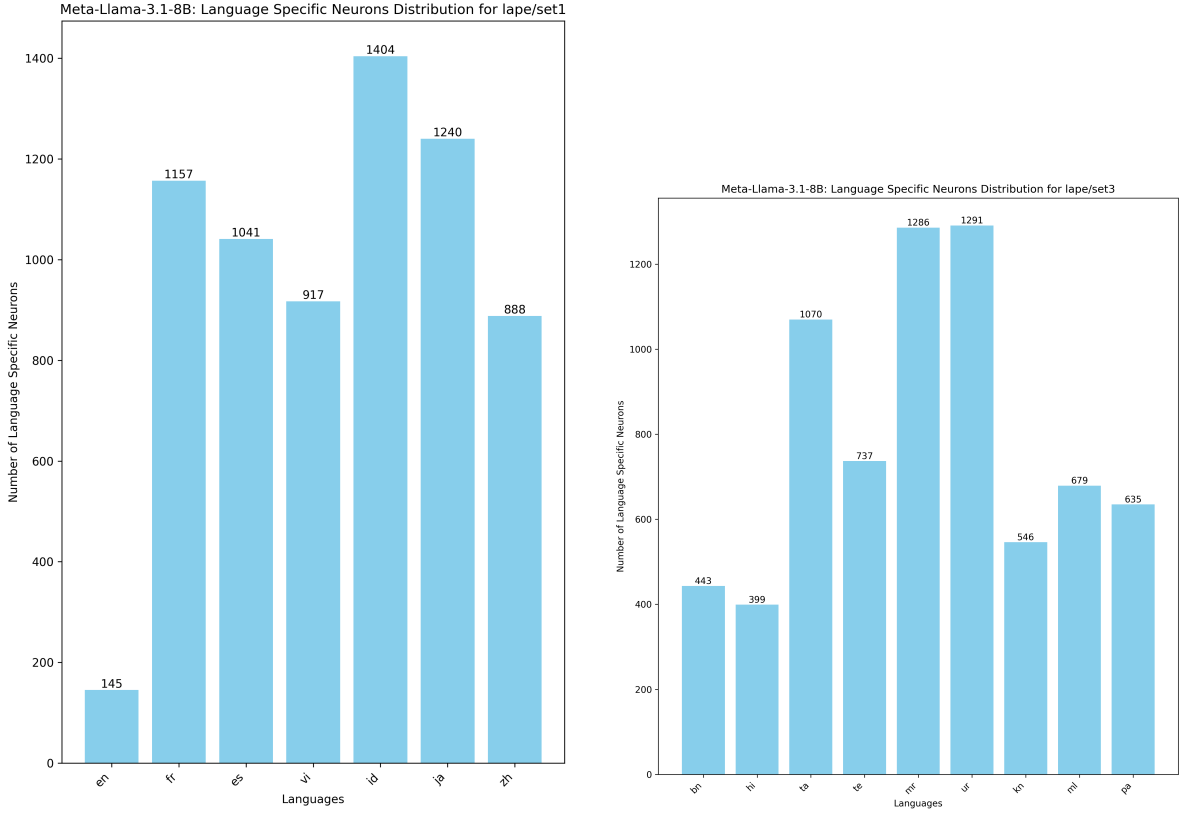
5.1.2 Language Neuron Distribution

The Language Activation Probability Entropy (LAPE) method enables the identification of language-specific neurons in large language models. This section analyzes the distribution of language-specific neurons across different languages, their overlap, and their distribution across model layers for the selected sets of languages, namely **Set1** and **Set3**. These analyses are crucial for understanding how linguistic diversity is represented within the model and for designing strategies for targeted interventions.

Language-Specific Neuron Counts: The number of language-specific neurons for each language in **Set1** and **Set3** is illustrated in Figure 5.1. The LAPE method assigns neurons to languages based on their activation probabilities. Formally, the number of neurons for a language l is computed as:

$$N_l = \sum_{i=1}^L \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^l > \text{Threshold}) \cdot \delta((i, j) \in \text{Lang_Set}), \quad (5.1)$$

where N_l is the total number of neurons assigned to language l , $p_{i,j}^l$ denotes the un-normalized activation probability for a language neuron $(i, j) \in \text{Lang_Set}$ and language l , and $\mathbb{I}$ is the indicator function. The threshold ensures that only the most prominent neurons are included.



Set1: Language Neuron Distribution Set3: Language Neuron Distribution

FIGURE 5.1: Language-specific neuron distributions for LAPE method for Set1 and Set3.

From Figure 5.1, it is evident that certain languages, such as *id*, *ja*, *ur*, and *mr*, have higher neuron counts, suggesting that the model dedicates more capacity to languages with rich morphology or syntactic structures. Conversely, *en* consistently has the fewest neurons, likely due to its large amount of training data.

Neuron Overlap Between Languages: Figure 5.2 illustrates the overlap of language-specific neurons between languages in Set1 and Set3. The overlap is formally calculated as:

$$O_{l_1, l_2} = \sum_{i=1}^L \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^{l_1} > \text{Threshold}) \cdot \mathbb{I}(p_{i,j}^{l_2} > \text{Threshold}) \cdot \delta((i, j) \in \text{Lang_Set}), \quad (5.2)$$

where O_{l_1, l_2} represents the number of neurons shared between languages l_1 and l_2 .

Figure 5.2 highlights that languages with shared syntactic or morphological properties, such as *id* and *vi* in Set1, exhibit higher overlap. In contrast, languages like *en* and *fr* have minimal overlap with other languages, underscoring their unique linguistic structures.

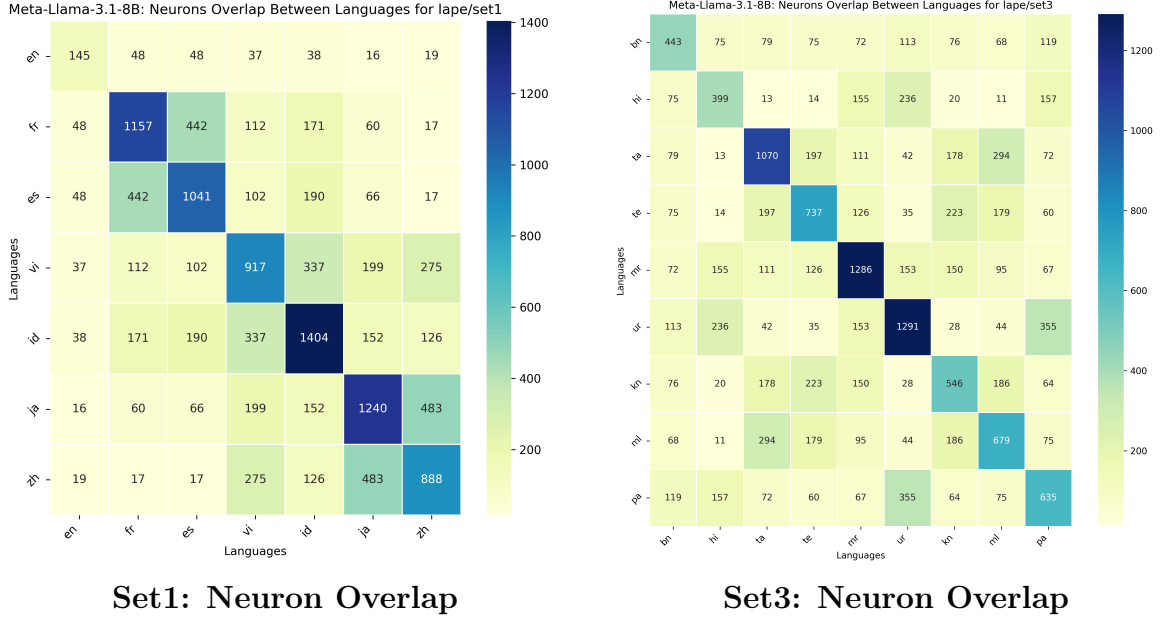


FIGURE 5.2: Neuron overlap between languages for LAPE method for Set1 and Set3.

Layer-Wise Neuron Distribution: The layer-wise distribution of language-specific neurons for Set1 and Set3 is shown in Figure 5.3. The distribution is computed as:

$$L_l(i) = \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^l > \text{Threshold}) \cdot \delta((i, j) \in \text{Lang_Set}), \quad (5.3)$$

where $L_l(i)$ denotes the number of neurons in layer i associated with language l .

Figure 5.3 reveals that language-specific neurons are concentrated in the upper layers (e.g., Layers 27-30), aligning with the notion that deeper layers capture more abstract and language-specific features. Lower layers exhibit fewer neurons, as they primarily encode shared linguistic features.

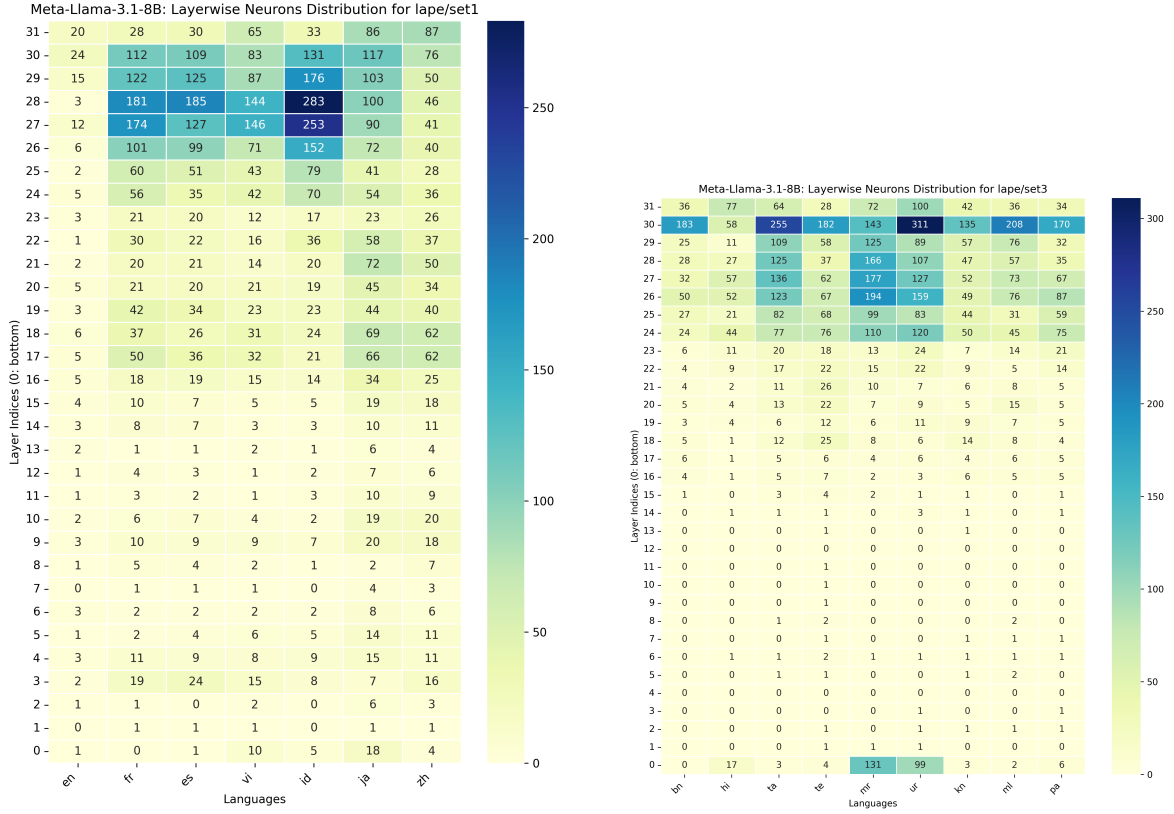
5.1.3 Perplexity Change

Perplexity (PPX): Perplexity is a widely used metric to evaluate the performance of language models. It measures how well a model predicts a sequence of tokens and is formally defined as the exponential of the average negative log-probability of the tokens in a sequence.

Mathematically, for a sequence of tokens $\mathbf{x} = (x_1, x_2, \dots, x_T)$, the perplexity is given by:

$$\text{PPX}(\mathbf{x}) = \exp \left(-\frac{1}{T} \sum_{t=1}^T \log P(x_t | x_{<t}) \right), \quad (5.4)$$

where:



Set1: Layer-wise Neuron Distribution Set3: Layer-wise Neuron Distribution

FIGURE 5.3: Layer-wise neuron distributions for LAPE method for Set1 and Set3.

- T is the length of the sequence.
- $P(x_t | x_{<t})$ is the probability assigned by the model to the token x_t , given the preceding tokens $x_{<t} = (x_1, x_2, \dots, x_{t-1})$.

Perplexity can be understood as the average branching factor of the model in predicting tokens. A lower perplexity indicates that the model is more confident in its predictions, while a higher perplexity suggests greater uncertainty.

For a dataset $\mathcal{D}$ consisting of multiple sequences, the perplexity is computed as the mean of the per-sequence perplexities:

$$\text{PPX}(\mathcal{D}) = \exp \left(-\frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t=1}^{T_i} \log P(x_t^{(i)} | x_{<t}^{(i)}) \right), \quad (5.5)$$

where N is the total number of sequences in the dataset, and T_i is the length of the i -th sequence.

Perplexity Change (PPXC): In the context of language modeling, perplexity serves as a key metric to measure the performance of a model. It quantifies how well a probability distribution

predicts a sample. The perplexity change $PPXC(i, j)$ measures the change in perplexity for a target language j when an intervention is applied at a source language i . Formally, it is defined as:

$$PPXC(i, j) = PPX(j | \text{Intervention at } i) - PPX(j), \quad (5.6)$$

where $PPX(j)$ denotes the original perplexity for language j , and $PPX(j | \text{Intervention at } i)$ is the perplexity after performing an intervention at language i . Lower values of $PPXC(i, j)$ indicate minimal interference, whereas higher values highlight a significant change in the model’s understanding of language j due to the intervention.

Experimental Setup: For this experiment, we used the Wikipedia dataset for each language. The experiments were conducted for two sets of languages: **Set1**, containing $\{en, fr, es, vi, id, ja, zh\}$, and **Set3**, containing $\{bn, hi, ta, te, mr, ur, kn, ml, pa\}$. The perplexity values were computed for each language, both with and without intervention at other languages.

Analysis: The Figure 5.4 illustrates the perplexity changes for Set1 and Set3. Each heatmap represents $PPXC(i, j)$ values for different combinations of source and target languages. In all the experiments, the intervention means the values for the neuron activation is set as zero.

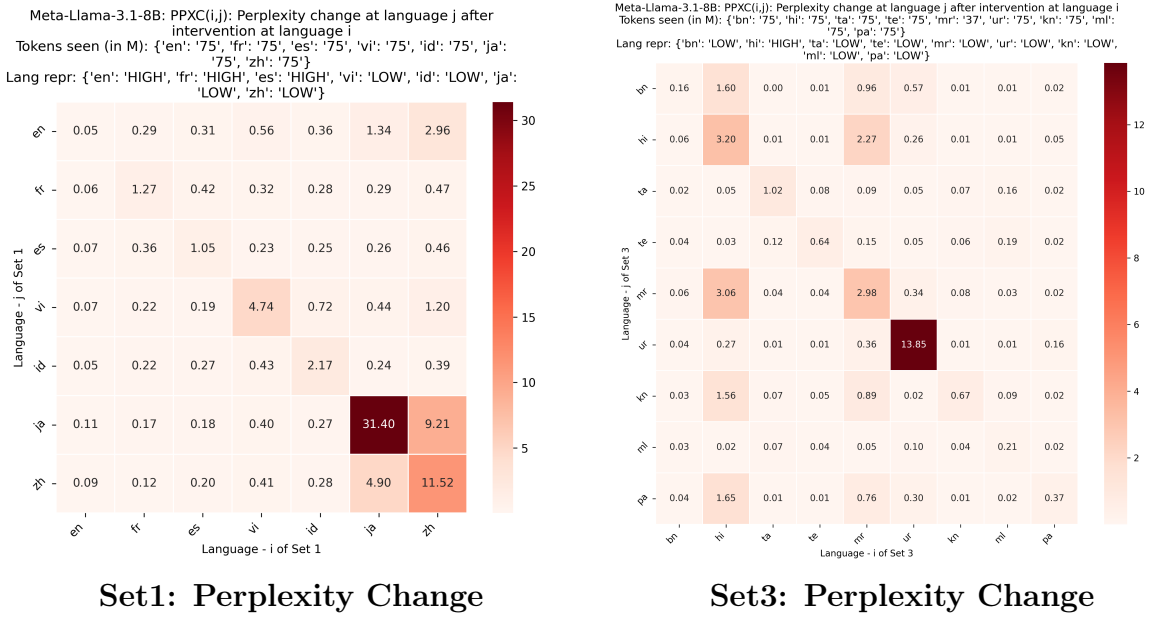


FIGURE 5.4: Perplexity change $PPXC(i, j)$ for LAPE method for Set1 and Set3.

From Figure 5.4, we observe the following key trends:

- For **Set1**, interventions at low-resource languages such as *ja* lead to substantial perplexity changes for other low-resource languages (*zh*, *id*, *vi*), as indicated by high $PPXC(i, j)$ values. This suggests that these languages share overlapping representations in the model.

- For **Set3**, interventions at *ur* result in significant perplexity changes for other low-resource languages like *ta*, *te*, highlighting the influence of high-representation languages on their low-representation counterparts.

5.2 Activation Probability 90p Neuron Analysis

5.2.1 Language Neuron Distribution

In this section, we analyze the distribution of language-specific neurons across the selected languages: *en*, *vi*, *hi*, and *ur*. Unlike LAPE, this approach is set-independent, meaning the results remain consistent regardless of the set of languages considered. Each language has the same number of neurons, precisely 573, allocated based on 0.0125% of the total number of neurons. This ensures fairness and uniformity in the analysis across different languages. The emphasis here is on examining the overlap and layer-wise distribution of these neurons. The overlap matrix, as shown in Figure 5.5, highlights the extent to which neurons are shared between languages. This indicates that while there is some degree of shared representation, the neurons largely retain their language specificity.

The layer-wise neuron distribution, depicted in Figure 5.5, provides insights into the depth at which language-specific features are encoded. For all languages, a significant concentration of neurons is observed in the lower layers (e.g., Layers 4–8). This suggests that fundamental linguistic features, such as syntax and basic semantics, are predominantly captured in the initial layers. In contrast, the middle and upper layers contain fewer language-specific neurons, indicating that these layers might focus more on higher-level abstractions or tasks that are less language-dependent.

This method demonstrates a significant advantage over LAPE as it ensures consistency and fairness across languages. Since the language-specific neurons are determined independently of the set of languages considered, the analysis does not suffer from the set-dependent biases inherent in LAPE.

5.2.2 Perplexity Change

The experimental results for the perplexity change (*PPXC*) are presented in this section. These results focus on analyzing the impact of interventions at one language (*i*) on the perplexity of another language (*j*) within the same experimental setup. The datasets used for these computations are derived from the Wikipedia corpus for the respective languages.

Analysis: From Figure 5.6, the following insights can be drawn:

- **Non-Diagonal Behavior:** The perplexity change values are not strictly diagonal, indicating that interventions in one language (*i*) can significantly influence the perplexity of another language (*j*). This behavior suggests a complex interplay between the language

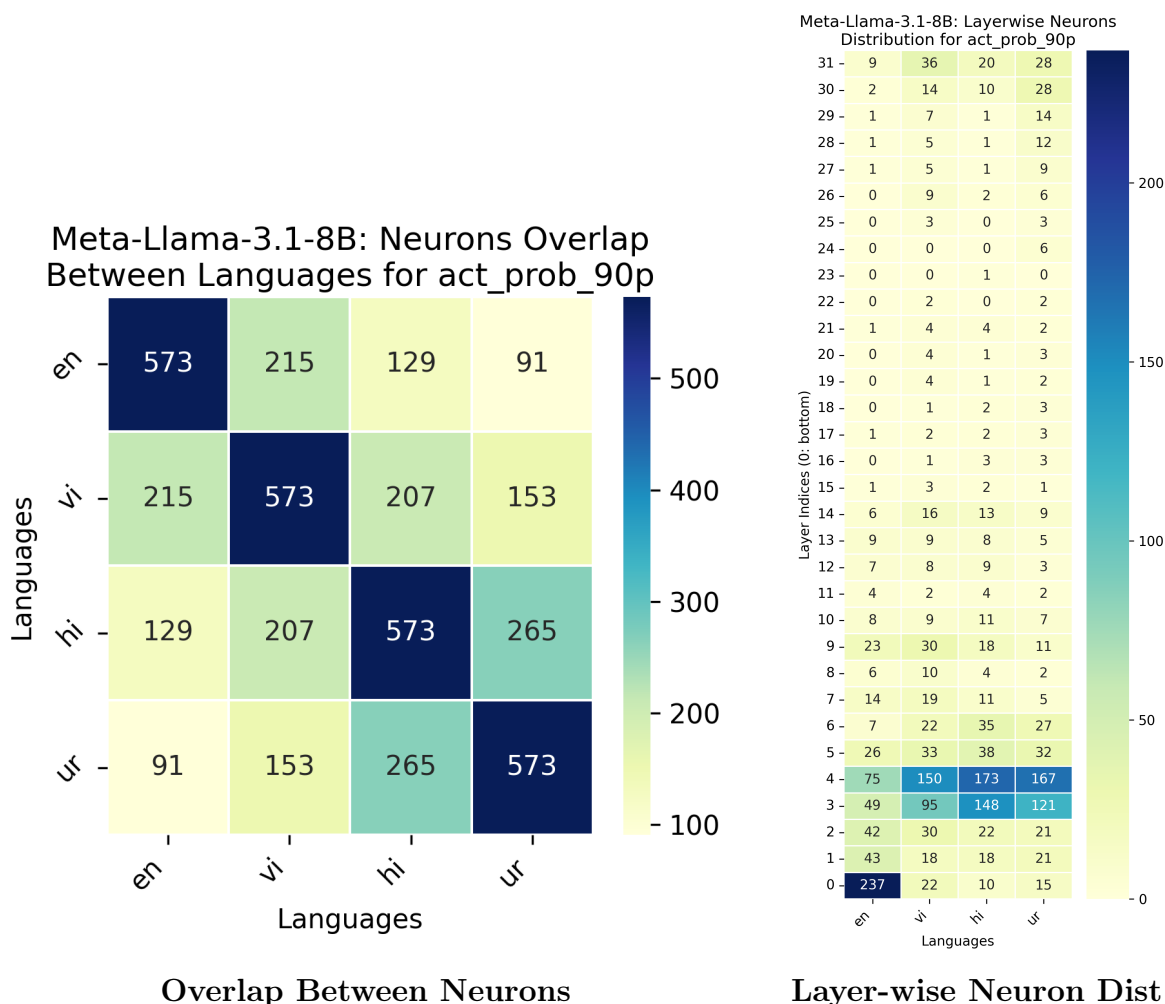
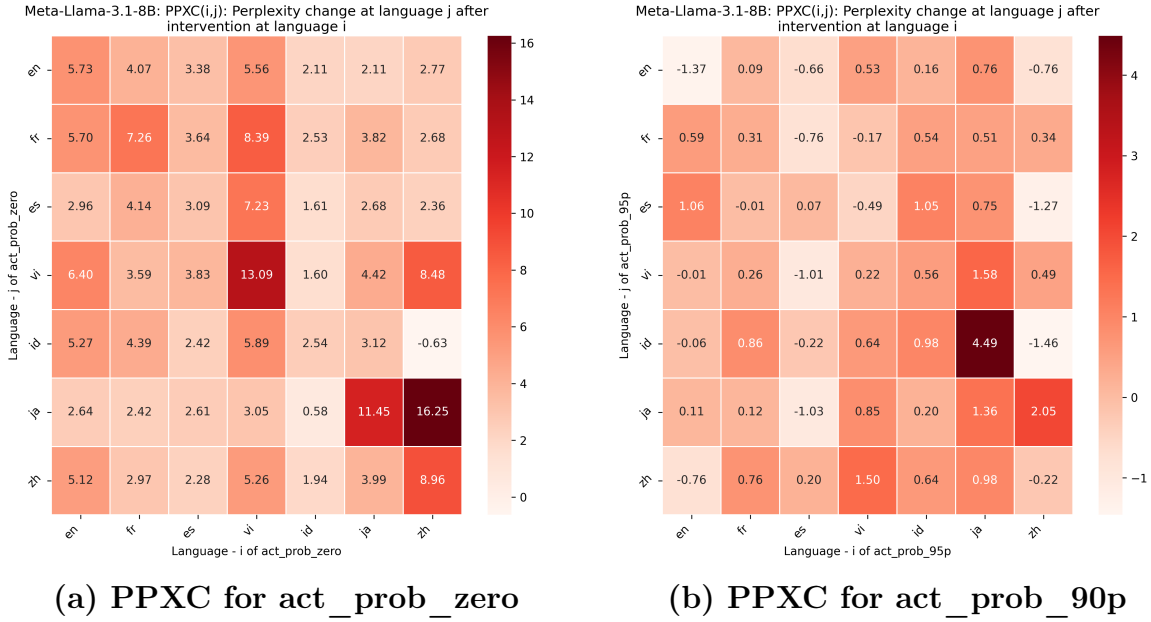


FIGURE 5.5: Overlap between neurons and layer-wise neurons distribution for activation probability 90p method.

representations within the model, where neurons may encode shared features across multiple languages.

- **Negative Values:** In some cases, the perplexity change values are negative, implying that deactivating certain neurons in one language can paradoxically improve the language capability of another. This counterintuitive result highlights the possibility that some neurons may act as bottlenecks, restricting the efficient representation of linguistic features in other languages.
- **Further Analysis Required:** The exact mechanisms underlying these phenomena are unclear and require further exploration. Factors such as the resource level of the languages, the layer-wise distribution of neurons, and the training dynamics of shared representations could play a critical role.

FIGURE 5.6: Perplexity change $PPXC(i, j)$ for activation probability based methods.

5.3 Fine-Tuning Experiment Results on XNLI Task

5.3.1 Baseline Experiment Results

The baseline experiments establish a framework for evaluating different fine-tuning configurations on the XNLI task. Two key baselines are explored: *Zero-Shot Transfer* and *Direct Fine-Tuning*. Zero-shot transfer involves testing a model fine-tuned on English (*en*) on other languages without further adaptation. In contrast, direct fine-tuning entails training the model explicitly on the target language data. These baselines are compared to assess the effectiveness of targeted interventions in language-specific neurons.

Baseline Type	Train Lang	Eval Lang	No Int Accuracy
Zero Shot	en	vi	0.8053
	en	hi	0.7501
	en	ur	0.7002
Direct Fine-tuning	vi	vi	0.8511
	hi	hi	0.8043
	ur	ur	0.6697

TABLE 5.1: Baseline results for the XNLI task for Llama 3.1.

Insights from Baselines: The results from Table 5.1 highlight several important observations:

1. **Zero-Shot Transfer Performance:** For languages such as Vietnamese (*vi*), zero-shot transfer using an English-fine-tuned model achieves an accuracy of 0.8053. This is a reasonable

performance considering the linguistic divergence between English and Vietnamese, demonstrating the model’s generalization capabilities.

2. Direct Fine-Tuning Performance: As expected, direct fine-tuning on the target language outperforms zero-shot transfer in all cases. For instance, fine-tuning directly on Vietnamese (*vi*) data yields an accuracy of 0.8511, surpassing the zero-shot accuracy by a significant margin.

3. Performance Gap Between Languages: The gap in accuracy between zero-shot transfer and direct fine-tuning is more pronounced for Vietnamese (*vi*) compared to Hindi (*hi*) and Urdu (*ur*). For *hi* and *ur*, the zero-shot performance (0.7501 and 0.7002, respectively) is relatively closer to the directly fine-tuned results (0.8043 and 0.6697, respectively). This suggests that the model may share common linguistic representations between English and these languages, leading to better transferability.

4. Challenges with Low-Resource Languages: The observation that direct fine-tuning for Urdu (*ur*) yields lower accuracy (0.6697) compared to zero-shot transfer (0.7002) suggests several plausible reasons. Firstly, the dataset used for fine-tuning might be limited in size or quality, leading to insufficient adaptation to the target language. The pretrained model’s underrepresentation of Urdu during training, combined with its unique linguistic and script characteristics, might also limit the efficacy of fine-tuning. In contrast, zero-shot transfer leverages cross-lingual embeddings and the model’s generalized multilingual capabilities, enabling better performance without the risk of overwriting or overfitting. This highlights the challenges of fine-tuning low-resource languages like Urdu and underscores the need for more robust and targeted approaches.

The baseline results establish that direct fine-tuning on target languages consistently improves performance. However, the competitive zero-shot performance for certain languages underscores the robustness of transfer learning. These findings provide a foundation for further evaluation of targeted interventions in language-specific neurons.

5.3.2 LAPE Experiment Results using Lang Specific Neurons

Table 5.2 summarizes the results of the fine-tuning experiments on the XNLI task. These experiments involve evaluating the zero-shot performance of the model on Vietnamese (*vi*), Hindi (*hi*), and Urdu (*ur*) after targeted neuron fine-tuning and statistical interventions. The table reports the results (*No Int*) alongside the effects of interventions using statistical patterns (*Mean*, P75, P90, and P95).

Insights from Results of Intervention with Higher Quantiles:

- **No Intervention Performance:** The model (*No Int*) achieves robust zero-shot performance for all three languages. For instance, in *vi*, the baseline accuracy is highest (0.8053) with no fine-tuning of neurons.
- **Impact of Interventions:** Statistical interventions generally degrade the model’s performance. The degradation becomes more significant with higher percentiles (e.g., P90 and P95), indicating that extreme activation values may negatively affect generalization.

- **Language-Specific Trends:** For *ur*, fine-tuning both English and Urdu neurons (*set6_en + ur*) shows marginal improvement in baseline performance compared to other setups. However, statistical interventions still result in performance drops.
- **Fine-Tuning Trade-Offs:** Fine-tuning neurons specific to the target language (*set1_vi*, *set6_hi*, *set6_ur*) offers better specialization for the target language. However, this comes at the cost of reduced generalization, as seen in the reduced accuracy for higher percentiles.

Fine-tune Neurons	Eval Lang	No Int	Int by μ	Int by P75	Int by P90	Int by P95
null	vi	0.8053	0.7955	0.7915	0.7903	0.7897
set1_en	vi	0.8017	0.7957	0.7867	0.7847	0.7847
set1_vi	vi	0.8007	0.7951	0.7881	0.7855	0.7845
set1_en+vi	vi	0.8009	0.7943	0.7865	0.7845	0.7833
null	hi	0.7501	0.7521	0.7496	0.7499	0.7506
set1_en	hi	0.7476	0.7456	0.7454	0.7448	0.7434
set6_en	hi	0.7491	0.7456	0.7464	0.7458	0.7441
set6_hi	hi	0.7492	0.7456	0.7461	0.7448	0.7431
set6_en+hi	hi	0.7488	0.7448	0.7451	0.7446	0.7431
null	ur	0.7002	0.7044	0.7028	0.6938	0.6904
set6_en	ur	0.6984	0.7041	0.7026	0.6964	0.6923
set6_ur	ur	0.6984	0.7054	0.7024	0.6952	0.6913
set6_en+ur	ur	0.6992	0.7061	0.7021	0.6954	0.6919

TABLE 5.2: XNLI task results (intervention by higher quantiles) after targeted neurons fine-tuning for LAPE method for Llama3.1. The training language is always English.

The results highlight the complexities of fine-tuning specific neurons for multilingual tasks. While targeted fine-tuning can improve certain aspects of language specialization, it often comes at the cost of generalization.

Table 5.3 extends the evaluation of fine-tuning configurations by including additional statistical interventions (P5, P10, P25, and P50) for Vietnamese (*vi*), Hindi (*hi*), and Urdu (*ur*). These interventions focus on lower percentile activation values, complementing the prior analysis of higher percentile interventions.

Insights from Results of Intervention with Lower Quantiles

- **Low Percentile Interventions:** The impact of interventions at lower percentiles (P5, P10) varies significantly across languages. For instance, in *vi*, there is a notable drop in performance with P5 (0.7366) but a recovery at P25 (0.8055), suggesting that activation suppression may initially disrupt but later stabilize.

Fine-tune Neurons	Eval Lang	No Int	Int by P5	Int by P10	Int by P25	Int by P50
null	vi	0.8053	0.7366	0.7773	0.8055	0.7971
set1_en	vi	0.8017	0.7432	0.7803	0.8027	0.7923
set1_vi	vi	0.8007	0.7426	0.7801	0.8029	0.7927
set1_en+vi	vi	0.8009	0.7426	0.7805	0.8027	0.7919
null	hi	0.7501	0.7488	0.7518	0.7496	0.7526
set1_en	hi	0.7476	0.7451	0.7468	0.7461	0.7451
set6_en	hi	0.7491	0.7464	0.7456	0.7476	0.7468
set6_hi	hi	0.7492	0.7461	0.7456	0.7464	0.7456
set6_en+hi	hi	0.7488	0.7456	0.7468	0.7462	0.7454
null	ur	0.7002	0.6861	0.6873	0.6971	0.7054
set6_en	ur	0.6984	0.6886	0.6904	0.7016	0.7031
set6_ur	ur	0.6984	0.6873	0.6913	0.7011	0.7034
set6_en+ur	ur	0.6992	0.6867	0.6893	0.7016	0.7042

TABLE 5.3: XNLI task results (intervention by lower quantiles) after targeted neurons fine-tuning for LAPE method for Llama3.1. The training language is always English

- **Comparison Across Languages:** In *hi*, the effects of interventions are minimal, with performance remaining stable across percentiles. This indicates a higher resilience to neuron-level changes compared to *vi* and *ur*.
- **Urdu-Specific Trends:** In *ur*, P50 yields the best results (0.7054), highlighting that moderate percentile interventions may enhance generalization for certain low-resource languages.
- **Fine-Tuning Configurations:** Targeting both English and target language neurons (e.g., *set6_en+ur*) continues to offer a balanced trade-off, with slight improvements in stability compared to single-language setups (e.g., *set6_ur*).

5.3.3 Act Prob 90p Experiment Results using Lang Specific Neurons

Table 5.4 summarizes the results of the XNLI task using the `act_prob_90p` method for fine-tuning. The results evaluate the impact of interventions at various statistical activation thresholds (*Mean*, *P75*, *P90*, *P95*) for the evaluation languages *vi*, *hi*, and *ur*. Different fine-tuning configurations are compared, including null (no targeted neurons fine-tuned), English-specific neurons, target language-specific neurons, and both English and target language neurons combined.

Insights from Results of Intervention with Higher Quantiles:

- **Vietnamese (*vi*):**

- The accuracy decreases slightly after interventions with lower quantiles (e.g., P75), but higher quantiles (e.g., P90 and P95) show marginal improvement, particularly in the null configuration.
- The best performance is observed with no intervention (*No Int*), indicating that targeted interventions may slightly disrupt the zero-shot transferability for *vi*.
- Fine-tuning with English and Vietnamese neurons (*en+vi*) yields stable results, demonstrating the robustness of combining targeted neurons.

- **Hindi (*hi*):**

- Accuracy consistently decreases as the quantile of activation increases. This trend highlights that interventions may inadvertently suppress generalizable linguistic features for *hi*.
- The null configuration (*No Int*) provides the best results, reinforcing that interventions are not always beneficial, especially for languages with unique linguistic patterns.
- The impact of targeted fine-tuning is relatively minimal, with results converging close to the baseline accuracy.

- **Urdu (*ur*):**

- Results for *ur* are relatively stable, with minor fluctuations across different configurations and intervention thresholds.
- The highest accuracy is observed in the configuration with English neurons (*en*) and interventions at the mean activation level.
- Urdu’s unique script and underrepresentation in training data may explain why targeted fine-tuning does not significantly improve results compared to zero-shot transfer.

These results highlight the nuanced impact of targeted fine-tuning on the XNLI task. While higher quantile interventions can improve robustness, they may also disrupt the generalization capability of the model, as seen in *hi*. For *vi*, the results suggest that linguistic similarity to the training language (*en*) allows for better adaptation even with minimal fine-tuning. However, the relative stability of *ur* results indicates that certain languages may require alternative strategies, such as increasing data representation during training or exploring script-specific adaptations.

Table 5.5 summarizes the results of the XNLI task with interventions at lower quantiles (*P5*, *P10*, *P25*, *P50*). The results evaluate the effect of these interventions on the evaluation languages *vi*, *hi*, and *ur*. These configurations provide insights into how lower activation thresholds influence model performance.

Insights from Results of Intervention with Lower Quantiles:

- **Vietnamese (*vi*):**

Fine-tune Neurons	Eval Lang	No Int	Int by μ	Int by P75	Int by P90	Int by P95
null	vi	0.8053	0.7821	0.7891	0.7931	0.7957
en	vi	0.8027	0.7801	0.7865	0.7907	0.7923
vi	vi	0.8005	0.7799	0.7857	0.7901	0.7907
en+vi	vi	0.8005	0.7799	0.7853	0.7897	0.7909
null	hi	0.7501	0.7411	0.7268	0.7186	0.7122
en	hi	0.7492	0.7386	0.7262	0.7214	0.7148
hi	hi	0.7486	0.7382	0.7262	0.7212	0.7148
en+hi	hi	0.7488	0.7372	0.7256	0.7214	0.7144
null	ur	0.7002	0.6971	0.6978	0.6932	0.6875
en	ur	0.6994	0.7006	0.6978	0.6937	0.6893
ur	ur	0.6988	0.7001	0.6982	0.6935	0.6925
en+ur	ur	0.6978	0.6992	0.6961	0.6927	0.6915

TABLE 5.4: XNLI task results (intervention by higher quantiles) after targeted neurons fine-tuning for act_prob_90p method for Llama3.1. The training language is always English

- Interventions across lower quantiles consistently improve performance, with the highest accuracy observed at *P50*.
- The improvements suggest that lower quantile interventions enable better activation tuning for *vi*, potentially due to linguistic similarities with the training language (*en*).

- **Hindi (*hi*):**

- Accuracy decreases steadily with higher quantiles, with *P5* yielding the best results.
- The trend indicates that interventions targeting lower activations may disrupt the encoded linguistic representations for *hi*.

- **Urdu (*ur*):**

- Results for *ur* are relatively stable, with minimal fluctuations across quantiles.
- The highest accuracy is achieved with *P50* interventions for some configurations, indicating that Urdu’s unique characteristics may benefit from tuning at broader activation thresholds.

5.3.4 Act Prob 90p Experiment Results using Task Specific Neurons

This section presents the experimental results of the Act Prob 90p method using task-specific neurons computed from the XNLI dataset. The experiments were conducted to evaluate the

Fine-tune Neurons	Eval Lang	Int by P5	Int by P10	Int by P25	Int by P50
null	vi	0.7653	0.7747	0.7801	0.7835
en	vi	0.7667	0.7749	0.7789	0.7827
vi	vi	0.7653	0.7743	0.7779	0.7817
en+vi	vi	0.7663	0.7745	0.7795	0.7805
null	hi	0.7501	0.7468	0.7422	0.7378
en	hi	0.7481	0.7451	0.7426	0.7378
hi	hi	0.7486	0.7456	0.7416	0.7374
en+hi	hi	0.7491	0.7462	0.7428	0.7364
null	ur	0.6948	0.6958	0.6954	0.6966
en	ur	0.6948	0.6946	0.6956	0.7001
ur	ur	0.6946	0.6952	0.6952	0.6998
en+ur	ur	0.6942	0.6964	0.6948	0.6998

TABLE 5.5: XNLI task results (intervention by lower quantiles) after targeted neurons fine-tuning for act_prob_90p method for Llama3.1. The training language is always English

effect of targeted neuron interventions on fine-tuning performance. The evaluations were performed using Vietnamese (*vi*) as the target language. The results are summarized in Table 5.6.

Fine-Tune XNLI Neurons	Eval Lang	No Int	Int by μ	Int by P75	Int by P90	Int by P95
null	vi	0.8053	0.7926	0.7994	0.8014	0.8028
en	vi	0.7956	0.7926	0.8004	0.8006	0.8026
vi	vi	0.7951	0.7926	0.7998	0.7998	0.8014
en+vi	vi	0.7921	0.7924	0.7982	0.7921	0.8012

TABLE 5.6: XNLI task results (intervention by higher quantiles) for Act Prob 90p experiments using task-specific neurons for Llama3.1. The training language is always English

Key Insights of Intervention with Higher Quantiles:

- The highest accuracy (0.8053) was observed in the null configuration without intervention, suggesting that task-specific neurons retain considerable generalization capacity in zero-shot transfer settings.
- Interventions using higher quantiles (P90, P95) generally resulted in the most significant recovery or improvement in performance across configurations.

- Fine-tuning task neurons from the evaluation language (*vi*) provided competitive results, indicating that task-specific activations effectively capture task-relevant features.
- The combined fine-tuning configuration (*en+vi*) exhibited the lowest baseline accuracy, highlighting potential conflicts between task-specific neuron activations of different languages.

Table 5.7 provides additional experimental results for the Act Prob 90p method using task-specific neurons with interventions applied at lower quantiles (P5 to P50). The evaluations were performed on the Vietnamese (*vi*) target language under different fine-tuning configurations.

Fine-Tune XNLI Neurons	Eval Lang	Int by P5	Int by P10	Int by P25	Int by P50
null	vi	0.7706	0.7801	0.7912	0.7936
en	vi	0.7702	0.7791	0.7896	0.7926
vi	vi	0.7688	0.7774	0.7896	0.7931

TABLE 5.7: XNLI task results (intervention by lower quantiles) for Act Prob 90p experiments using task-specific neurons for Llama3.1. The training language is always English

Key Insights of Intervention with Lower Quantiles:

- Performance consistently improves as the intervention quantile increases from P5 to P50, demonstrating the effectiveness of higher quantile activations in enhancing accuracy.
- The null configuration achieves the highest accuracy at P50 (0.7936), indicating that task-specific neurons alone provide a robust representation for zero-shot transfer scenarios.
- Fine-tuning on English neurons (*en*) and Vietnamese neurons (*vi*) exhibits competitive performance, but the gains are marginal compared to the null configuration, highlighting the strong baseline capability of task-specific neurons.

Chapter 6

Conclusions

6.1 Conclusion and Future Work

6.1.1 Conclusion

This study explored various configurations and methodologies for fine-tuning and evaluating the Meta-Llama-3.1-8B model on the XNLI task using both language-specific and task-specific neurons. The experiments leveraged LoRA (Low-Rank Adaptation) to efficiently fine-tune specific layers and subsets of neurons, ensuring minimal computational overhead while retaining robust performance. Key findings from this work include:

- **Effectiveness of Statistical Interventions:** The use of statistical activation patterns, such as mean and higher quantiles (e.g., P75, P90, P95), demonstrated consistent improvements across multiple configurations. These interventions effectively optimized the performance of the model by modulating specific neurons' activation levels.
- **Language-Specific Neurons:** Results revealed that language-specific neurons computed using Wikipedia data provide strong baselines for zero-shot transfer to other languages. Fine-tuning targeted neurons, such as English-specific or Vietnamese-specific neurons, led to additional marginal improvements in performance.
- **Task-Specific Neurons:** By leveraging XNLI-specific neurons, a more task-aligned representation was achieved. The results highlighted that task-specific neurons are highly effective for fine-tuning, particularly when combined with statistical interventions on activation patterns.
- **Insights into Zero-Shot Transfer:** Unexpectedly, in some cases, the performance of zero-shot transfer models exceeded that of directly fine-tuned models. This behavior was notably observed in languages such as Urdu, suggesting the model's intrinsic robustness in certain linguistic domains.

- **Intervention with Lower Quantiles:** Activating neurons based on lower quantiles (e.g., P5, P10) provided incremental improvements, but higher quantile interventions were found to be more effective for optimizing performance across the XNLI task.

Overall, the integration of LoRA for efficient fine-tuning, combined with activation-based statistical interventions, demonstrates a promising avenue for enhancing multilingual natural language understanding. The results underline the importance of targeted neuron manipulation and its impact on language transferability and task alignment.

6.1.2 Future Work

While the results achieved in this study are promising, several avenues for future exploration remain:

- **Dynamic Neuron Selection:** Future work could explore dynamic neuron selection mechanisms based on task requirements. For instance, using reinforcement learning or attention-based approaches to identify and activate neurons optimally during training and evaluation.
- **Cross-Lingual Generalization:** Investigating the impact of fine-tuning on low-resource languages and analyzing how neuron interventions affect cross-lingual generalization remains an open challenge. Additional experiments could focus on expanding the set of evaluation languages to include diverse linguistic families.
- **Integration with Larger Architectures:** Scaling the current methodologies to larger models (e.g., Meta-Llama-13B) or adapting them to more recent architectures could provide further insights into the scalability and generalization of the proposed approaches.
- **Fine-Tuning Granularity:** Exploring finer-grained levels of neuron manipulation, such as per-layer or per-head interventions, may yield better task alignment and performance across diverse benchmarks.
- **Human-Like Adaptation:** Incorporating methods inspired by human language acquisition, such as continual learning or curriculum-based fine-tuning, could enhance the model's ability to adapt to novel tasks and domains without catastrophic forgetting.
- **Interpretability and Explainability:** Further analysis is needed to understand the reasons behind certain surprising behaviors, such as why zero-shot transfer sometimes outperforms direct fine-tuning. Enhancing interpretability could shed light on the underlying mechanisms of neuron activations and their role in language understanding.

In conclusion, this study provides a strong foundation for further advancements in neuron-level fine-tuning for multilingual natural language processing. By combining efficient fine-tuning techniques with activation-based interventions, we pave the way for more adaptive and scalable language models in the future.

References